Google Cloud H2S

AI Builder Cup

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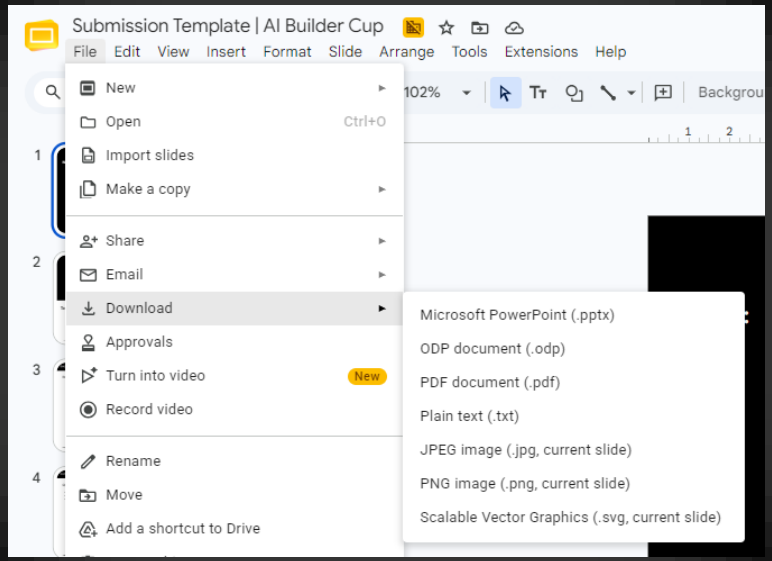


Team Details

Team name: Kisan Mitra

Team leader name: Kavita Chegaraddi

Problem Statement: Sustainability & Social Impact — Smallholder Crop Advisory Agent for tomato farmers in the Kolar belt, Karnataka



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Brief about the idea

- A smallholder tomato farmer in the Kolar belt, Karnataka spots a diseased leaf with no agronomist to call and unreliable connectivity to search with.
- Existing diagnosis apps (Plantix, PlantVillage) identify the disease but stop there — they don't say where to buy the treatment nearby, what it costs, or whether a nearby outbreak means the farmer should check sooner.
- Kisan Mitra turns a photo and a spoken description into a cited diagnosis, a spoken Kannada advisory, the nearest in-stock dealer with today's price, a proactive outbreak alert, a crop-loss insurance evidence packet, a group-buy discount alert, and an anonymized district dashboard for extension officers.
- Target users: smallholder tomato farmers in the Kolar tomato belt, Karnataka — typically 0.5–4 acre plots, variable literacy, intermittent connectivity, no dedicated access to an agronomist. Primary language: Kannada.

Opportunities

- How different is it: Plantix diagnoses, AgroStar connects to dealers, PlantVillage pairs AI with weather — none combine diagnosis, voice, and market linkage in one flow, and none reason across farmers or across time on the same plot.
- How it solves the problem: the outbreak alert reaches a farmer before their own crop shows damage; the insurance claim assistant turns a diagnosis into a fileable claim; the group-buy alert turns an early warning into a coordinated bulk discount; the district dashboard turns farmer-level data into a resourcing signal for an extension officer or NGO.
- USP: moves the system from advice to economic action and policy support — no existing tool in this space does all four.

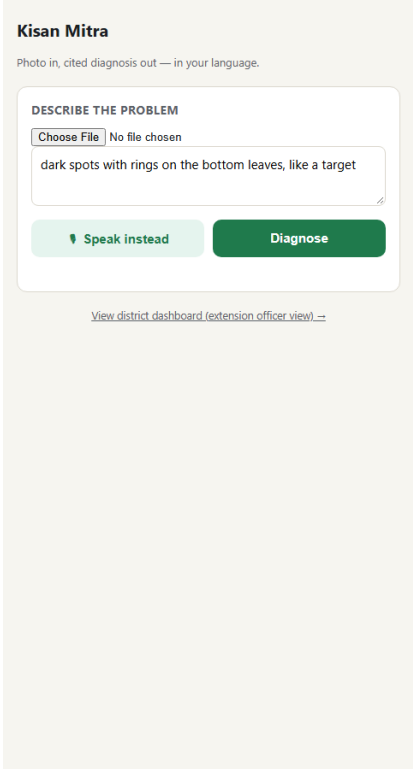
List of features offered by the solution

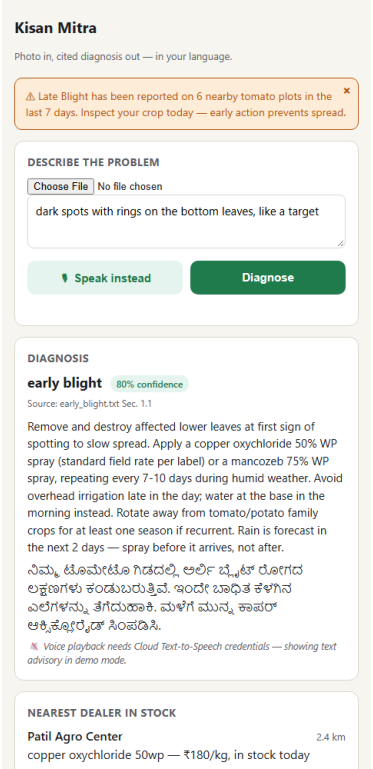
- Cited disease diagnosis from a photo + spoken description — not a black-box guess
- Spoken advisory in Kannada, weather-aware (holds off spraying before forecast rain)
- Nearest in-stock dealer lookup with today's price
- Proactive outbreak early-warning — alerts nearby farmers before their own crop shows symptoms
- Plot-level memory — flags recurring disease as a root-cause issue (e.g. drainage), not just a repeat treatment
- Sell-side market linkage — best nearby mandi price when a healthy crop is near harvest
- Crop-loss insurance claim assistant — auto-generates a fileable evidence packet
- Group-buy / collective bargaining alert — coordinates nearby farmers for a bulk discount
- District dashboard — anonymized, village-level view for an extension officer or NGO

Process flow diagram or Use-case diagram

- Farmer speaks/types symptom + uploads photo
- → Gemini multimodal diagnosis grounded via Vertex AI Search (RAG) over the regional disease guide
- → Weather tool call (Open-Meteo) checked before recommending a spray
- → Branch: diseased → dealer lookup + outbreak check + insurance claim check + group-buy check | healthy + near harvest → sell-price lookup
- → Advisory translated to Kannada, converted to speech
- → Response rendered: diagnosis, dealer card, outbreak banner, insurance packet, group-buy card
- → Separately: district dashboard aggregates all diagnosis history, anonymized to village level, for an extension officer

Wireframes / Mock diagrams of the proposed solution





Architecture diagram of the proposed solution

- Farmer (photo + voice) → Firebase Hosting (frontend) → Cloud Run (FastAPI backend)
- Gemini multimodal diagnosis, grounded via Vertex AI Search over the regional disease guide
- Cloud Translation + Cloud Text-to-Speech for the Kannada voice advisory
- Firestore holds the dealer directory, mandi prices, and diagnosis history
- Diagnosis history feeds three agentic reasoning steps: a geo-radius outbreak query, a per-plot history query for recurring issues, and a district-wide aggregation for the extension-officer dashboard
- An outbreak signal additionally triggers a bulk-pricing check against the dealer directory for the group-buy alert

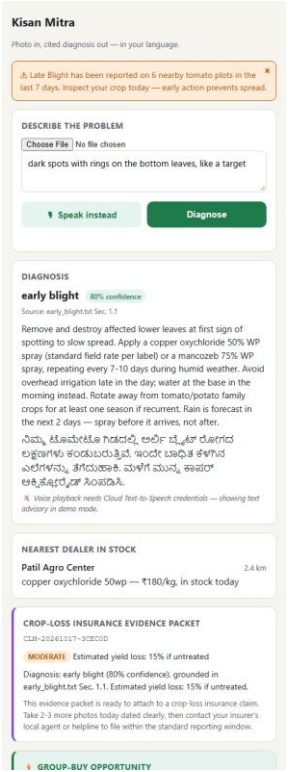
Technologies to be used in the solution

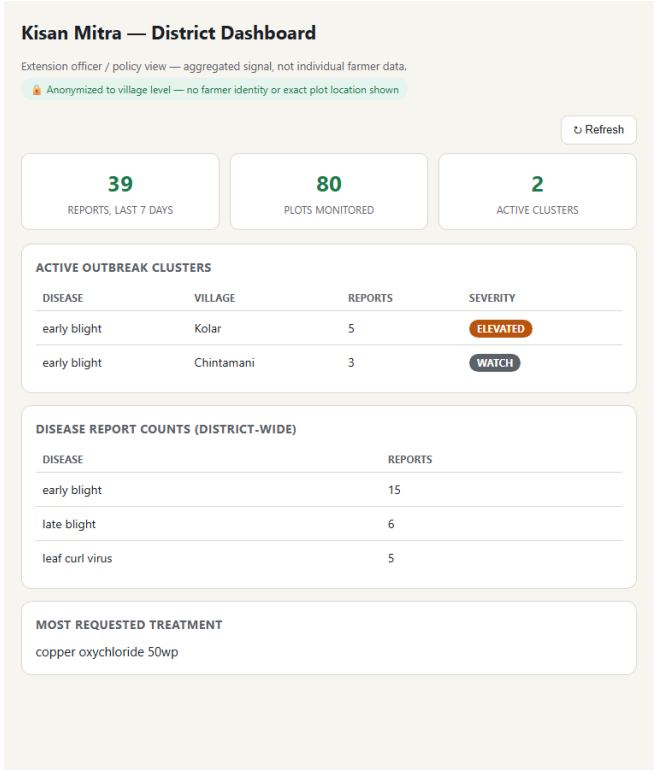
- Gemini (Vertex AI) — multimodal diagnosis, function calling, synthesis
- Vertex AI Search — RAG grounding over the regional disease guide
- Cloud Run — FastAPI backend hosting
- Firebase Hosting — frontend hosting
- Firestore — dealer directory, mandi prices, diagnosis history
- Cloud Translation API + Cloud Text-to-Speech — Kannada voice advisory
- Open-Meteo — live weather context
- Python (FastAPI, pydantic), scikit-learn (RAG retrieval), Playwright (verification)

Estimated implementation cost

- Gemini API (Vertex AI): pay-per-call, multimodal + text — estimated low cost at pilot scale (a few thousand diagnoses/month)
- Vertex AI Search: billed per query + storage — a single regional disease guide (9 documents) is a negligible index size
- Cloud Run + Firebase Hosting: both have a generous free tier; this workload stays within it at pilot scale
- Cloud Translation + Text-to-Speech: billed per character — Kannada advisory text is short (1–2 sentences per diagnosis)
- Firestore: free tier covers a pilot-scale dealer directory and diagnosis history
- Google Cloud's $300 new-account credit comfortably covers hackathon build and pilot testing

Snapshots of the prototype





Prototype Performance report / Benchmarking

- Measured live against the running FastAPI backend (5 requests per endpoint, mock mode):
- GET /health — avg 5ms
- GET /dealer-lookup — avg 9ms
- GET /outbreak-check — avg 30ms (geo-radius query over 300 diagnosis records)
- GET /district-dashboard — avg 24ms (full district aggregation, anonymized to village level)
- POST /diagnose (full pipeline incl. live Open-Meteo weather call) — avg 924ms
- Verified end-to-end with Playwright against the live browser UI — zero console errors across the main app and the district dashboard

Additional Details / Future Development

- Offline-first fallback (planned): a distilled on-device TensorFlow Lite classifier for the 8–10 most common regional diseases, so core diagnosis keeps working when connectivity drops entirely. Training script already written, pending a real multi-hour training run on the full PlantVillage dataset.
- Expansion path: adding a new crop or region means indexing a new disease guide into Vertex AI Search, not rewriting the diagnosis logic — the architecture is crop-agnostic by design even though this build is scoped to tomato in Karnataka for the demo.
- Resource & environmental impact: targeted treatment instead of blanket spraying, weather-timed application to reduce chemical runoff, and earlier detection to reduce crop loss — moving this beyond social impact into genuine sustainability.

Provide links to your:

GitHub Public Repository: [add link before submission]

Demo Video Link (3 Minutes): [add link before submission]

Final Product Link: [add link before submission]

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Thank you

